IT24100378 – Razni Ahamed M.R
Assigned Model: Support Vector Machine (SVM)

1. Individual Role and Contributions

- My main responsibility in the GrainWise project was developing and evaluating the Support Vector Machine (SVM) classification model to distinguish between Cammeo and Osmancik rice varieties using their physical measurements.
- I developed the SVM.ipynb notebook, implementing model training, hyperparameter tuning, repeated cross-validation, feature importance analysis, and stability testing. I also contributed to the project's practical demonstration by developing the web interface and prediction API, which allow users to enter rice grain measurements and obtain predictions from the final selected model.

2. Technical Learning and Key Decisions

- One of my main learning experiences was understanding how SVM identifies a decision boundary that separates different classes. I explored the Radial Basis Function (RBF) kernel and how regularisation and kernel parameters influence classification performance. I used GridSearchCV to evaluate 12 hyperparameter combinations, considering different values of C and gamma. The best configuration used C = 10, gamma = 0.01, and the RBF kernel.
- I also learned why feature standardisation is important for SVM, as features with larger numerical ranges can dominate distance-based kernel calculations. Using repeated stratified cross-validation, my model achieved 92.81% accuracy, 91.55% F1-score, and 97.99% ROC-AUC. These results helped me understand the importance of evaluating classification models using multiple performance metrics.

3. Challenges and Problem-Solving

- A major challenge was determining whether a nonlinear kernel was necessary for the dataset. I compared linear and RBF kernels and found that the linear SVM achieved approximately 92.95% accuracy, slightly higher than the tuned RBF model. This helped me understand that additional model complexity does not always improve performance.
- Another challenge was interpreting feature importance because RBF SVM does not provide directly interpretable coefficients. I used permutation importance to identify influential features, particularly Aspect_Ratio and Major_Axis_Length.

4. Critical Evaluation and Reflection

- Comparing SVM with Logistic Regression, KNN, and Random Forest helped me understand that different algorithms can achieve similar classification performance. Although SVM performed competitively, Logistic Regression was selected as the final model because of its computational efficiency, interpretability, and flexibility for cost-sensitive classification.
- I also evaluated SVM using different random seeds (42, 7, and 2024), obtaining consistent performance. This strengthened my understanding of model stability, reliable validation, and evidence-based model selection.

5. Personal Growth and Future Improvements

- This project improved my understanding of SVM, kernel selection, hyperparameter tuning, cross-validation, and model interpretation. Developing the prediction interface also helped me understand how machine learning models can be integrated into practical applications.
- Most importantly, I learned that selecting an appropriate model requires considering accuracy, interpretability, computational cost, and real-world requirements rather than focusing only on predictive performance. In future projects, I would explore additional kernel configurations, probability calibration, and testing with rice samples collected under different imaging conditions.